

# How AI Impacts Skill Formation

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## Abstract

AI assistance produces significant productivity gains across professional domains, particularly for novice workers. Yet how this assistance affects the development of skills required to effectively supervise AI remains unclear. Novice workers who rely heavily on AI to complete unfamiliar tasks may compromise their own skill acquisition in the process. We conduct randomized experiments to study how developers gained mastery of a new asynchronous programming library with and without the assistance of AI. We find that AI use impairs conceptual understanding, code reading, and debugging abilities, without delivering significant efficiency gains on average. Participants who fully delegated coding tasks showed some productivity improvements, but at the cost of learning the library. We identify six distinct AI interaction patterns, three of which involve cognitive engagement and preserve learning outcomes even when participants receive AI assistance. Our findings suggest that AI-enhanced productivity is not a shortcut to competence and AI assistance should be carefully adopted into workflows to preserve skill formation – particularly in safety-critical domains.

## 1 Introduction

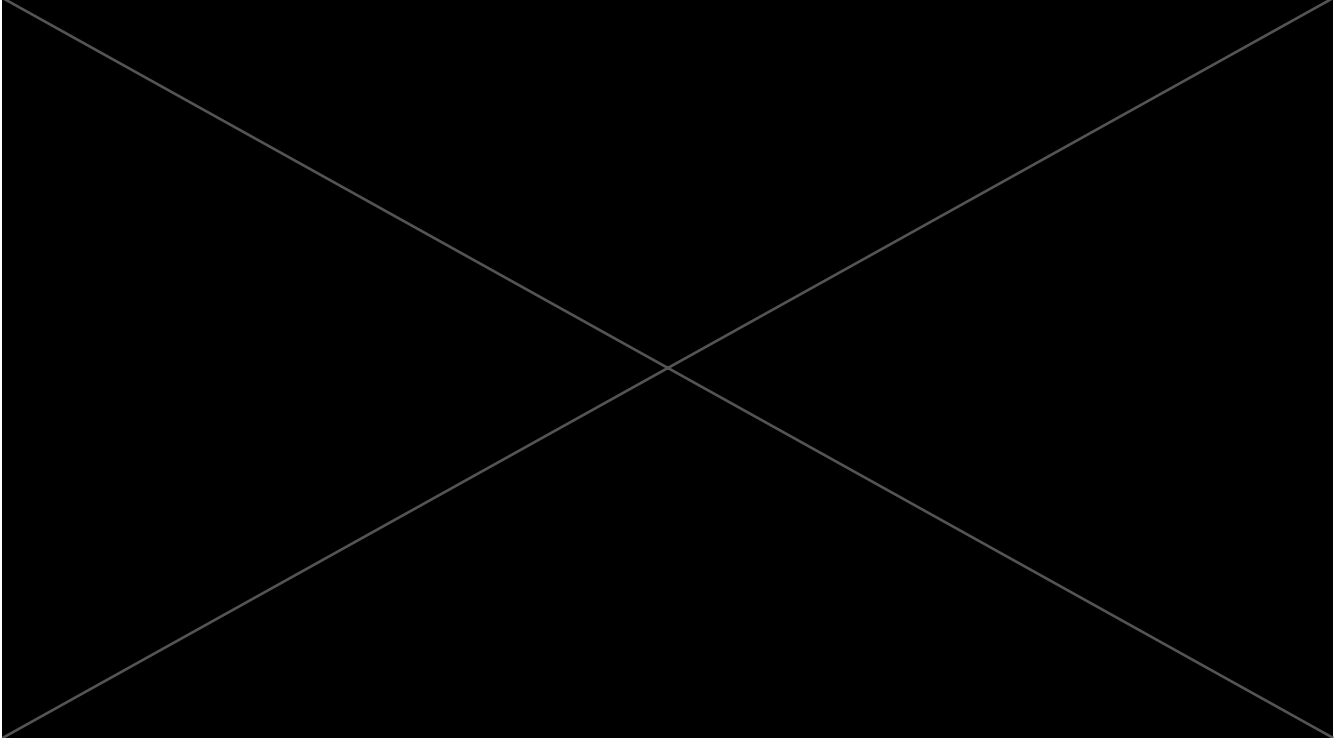
Since the industrial revolution, skills in the labor market have continually shifted in response to the introduction of new technology; the role of workers often shifts from performing the task to supervising the task [Autor et al., 2001]. For example, the automation of factory robots has enabled humans to move from manual labor to supervision, and accounting software has enabled professionals to move from performing raw calculations to identifying better bookkeeping and tax strategies. In both scenarios, humans are responsible for the quality of the final product and are liable for any errors [Bleher and Braun, 2022]. Even as automation changes the process of completing tasks, technical knowledge to identify and fix errors remains extremely important.

As AI promises to be a catalyst for automation and productivity in a wide range of applications, from software engineering to entrepreneurship [Dell’Acqua et al., 2023, Peng et al., 2023, Cui et al., 2024, Otis et al., 2024, Brynjolfsson et al., 2025], the impacts of AI on the labor force are not yet fully understood. Although more workers rely on AI to improve their productivity, it is unclear whether the use of AI assistance in the workplace might hinder core understanding of concepts or prevent the development of skills necessary to supervise automated tasks. Although most studies have focused on the end *product* of AI assistance (e.g., lines of code written, quality of ideas proposed), an equally important, if not more crucial question is how *process* of receiving AI assistance impacts workers. As humans rely on AI for skills such as brainstorming, writing, and general critical thinking, the development of these skills may be significantly altered depending on how AI assistance is used.

Software engineering, in particular, has been identified as a profession in which AI tools can be readily applied and AI assistance significantly improves productivity in daily tasks [Peng et al., 2023, Cui et al., 2024]. Junior or novice workers, in particular, benefit most from AI assistance when writing code. In high-stakes applications, AI written code may be debugged and tested by humans before a piece of software is ready for deployment. This additional verification that enhances safety is only possible when human engineers themselves have the skills to understand code and identify errors. As AI development progresses, the problem of supervising more and more capable AI systems becomes more difficult if humans have weaker abilities

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to understand code [Bowman et al., 2022]. When complex software tasks require human-AI collaboration, humans still need to understand the basic concepts of code development even if their software skills are complementary to the strengths of AI [Wang et al., 2020]. The combination of persistent competency requirements in high-stakes settings and demonstrated productivity gains from AI assistance makes software engineering an ideal testbed for studying how AI affects skill formation.

We investigate whether using and relying on AI affects the development of software engineering skills [Handa et al., 2025]. Based on the rapid adoption of AI for software engineering, we are motivated by the scenario of engineers acquiring new skills on the job. Although the use of AI tools may improve productivity for these engineers, would they also inhibit skill formation? More specifically, does an AI-assisted task completion workflow prevent engineers from gaining in-depth knowledge about the tools used to complete these tasks? We run randomized experiments that measure skill formation by asking participants to complete coding tasks with a new library that they have not used before. This represents one way in which engineers acquire and learn new skills, since new libraries are frequently introduced in languages such as Python. We then evaluate their competency with the new library. Our main research questions are (1) whether AI improves productivity for a coding task requiring new concepts and skills, and (2) whether this use of AI reduces the level of understanding of these new concepts and skills.

## 1.1 Our Results

Motivated by the salient setting of AI and software skills, we design a coding task and evaluation around a relatively new asynchronous Python library and conduct randomized experiments to understand the impact of AI assistance on task completion time and skill development. We find that using AI assistance to complete tasks that involve this new library resulted in a reduction in the evaluation score by 17% or two grade points (Cohen’s  $d = 0.738$ ,  $p = 0.010$ ). Meanwhile, we did not find a statistically significant acceleration in completion time with AI assistance (Figure 6).

Through an in-depth qualitative analysis where we watch the screen recordings of every participant in our main study, we explain the lack of AI productivity improvement through the additional time some participants invested in interacting with the AI assistant. Some participants asked up to 15 questions or spent more than 30% of the total available task time on composing queries (Figure 12). We attribute the gains in

**Overreliance** Although much of the literature in economics on AI-enhanced productivity implicitly assumes that generations of AI are trustworthy, the reality is that generative AI can produce incorrect [Longwell et al., 2024] or hallucinated content [Maleki et al., 2024]. When models are fallible, yet still deployed to assist humans, human decisions that follow erroneous model decisions are referred to as “overreliance” [Buçinca et al., 2021, Vasconcelos et al., 2023, Klingbeil et al., 2024]. Although methods have been suggested to reduce overreliance, these focus mainly on decision-time information such as explanations [Vasconcelos et al., 2023, Reingold et al., 2024] or debate [Kenton et al., 2024].

## 2.2 CS Education and AI Assistance

Measuring the acquisition of skills is highly domain dependent. For computer science in particular, most introductory courses measure learning through multiple choice questions, code writing, and code reading/explanations [Cheng et al., 2022]. More recent work has found code interviews, and active discussion of students’ code to yield positive learning outcomes [Kannam et al., 2025].

Several observational studies have described how students use AI tools in the context of a computer science course. [Poitras et al.] found, over the course of a semester, that students used AI tools to write code, fix errors, and explain algorithmic concepts; students with less coding proficiency were more likely to seek AI assistance. Other works use surveys to find that students may be hesitant to use AI coding assistant tools due to “dependence worry” (i.e., overreliance on coding tools) [Pan et al., 2024]. For formal methods, [Prasad et al.] coded the different ways in which students used LLMs for course work and found that upper-year students taking the class did not rely on LLM assistance and only asked a few questions at the beginning.

User studies have also been conducted in the professional development environments. [Wang et al.] study different patterns in usage between users with and without chat access to AI models in completing coding puzzles and development tasks. They found rich interaction patterns including interactive debugging, code discussions, and asking specific questions. Participants ranged from asking ChatGPT to do then the entire problem (lowest quality code output) to only asking minimal questions (highest efficiency). Other studies have reported that AI tools help the software development process through easier access to documentation and accurate generation code for specific APIs [Pinto et al., 2024].

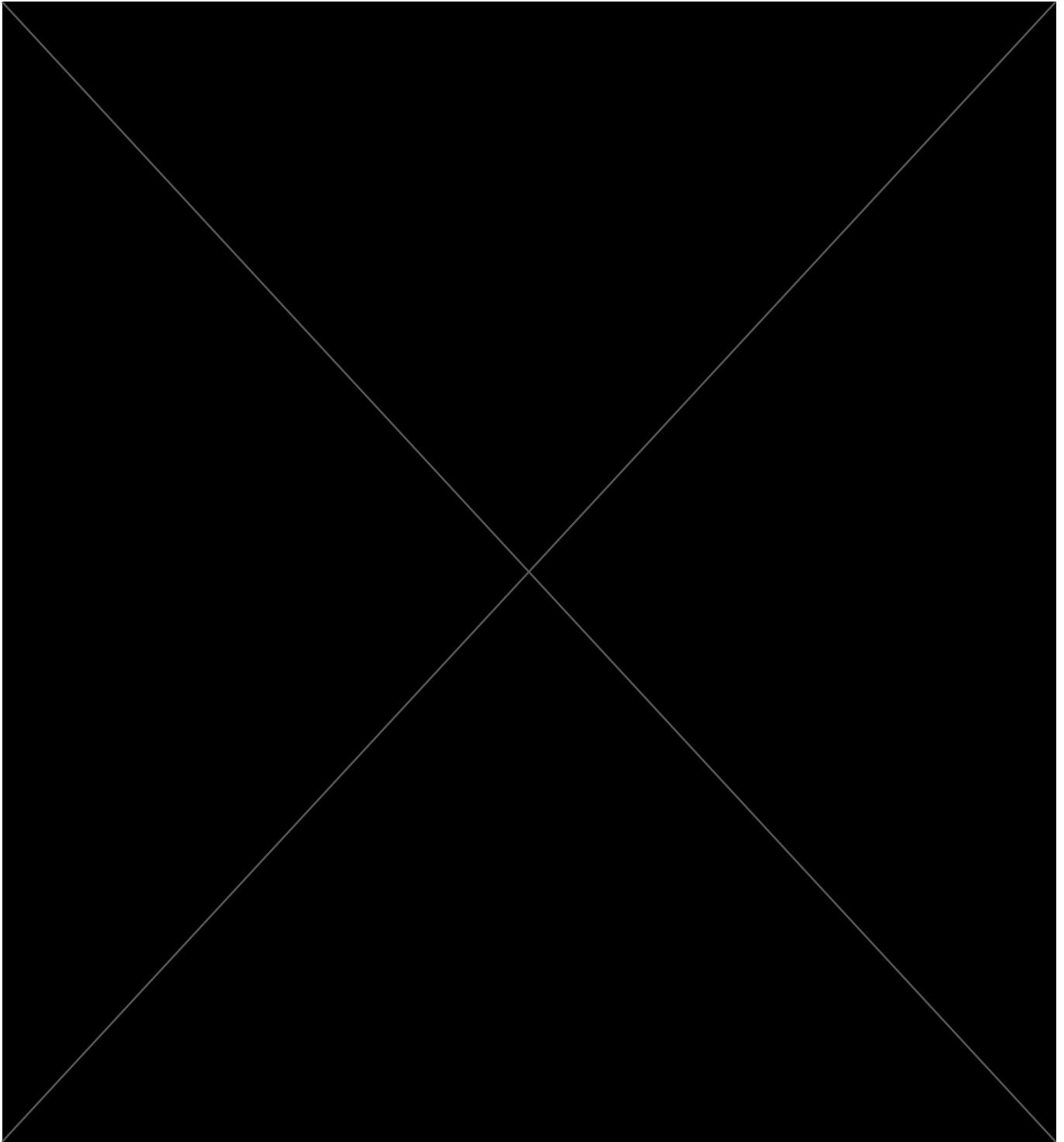
## 3 Framework

**Professional Skill Acquisition** The “learning by doing” philosophy has been suggested by many learning frameworks such as the Kolb’s experiential learning cycle, and the Problem-Based Learning (PBL) [Kolb, 2014, Schmidt, 1994]. The frameworks connect the completion of real-world tasks with the learning of new concepts and the development of new skills. Experiential learning has also been explored specifically in software engineering courses in higher education in order to mimic solving problems in a professional setting [Gonzalez-Huerta et al., 2020]. In its simplest form, we model AI tool assistance as taking a different learning path than without AI. We hypothesize that using AI tools to generate code in the development process effectively amounts to taking a shortcut to task completion without a pronounced learning stage.

**AI for Coding Usage Patterns** Prior works have found that humans use AI in many different ways for coding: from question answering to writing code, to debugging [Poitras et al., 2024, Wang et al., 2020, Pinto et al., 2024]. In our framework, different ways of using AI assistance represent different learning paths taken to reach the goals of completing the task. We analyze these different usage patterns in the qualitative analysis of this work (Section 6).

**Research Questions** Based on this background, we focus on on-the-job learning: settings where workers must acquire new skills to complete tasks. We seek to understand both the impact of AI on productivity and skill formation. We ask whether AI assistance presents a tradeoff between immediate productivity and longer-term skill development or if AI assistance presents a shortcut to enhance both. Our research questions are as follows:

- **RQ1:** Does AI assistance improve task completion productivity when new skills are required?



## 7 Discussion

Our main finding is that using AI to complete tasks that require a new skill (i.e., knowledge of a new Python library) reduces skill formation. In a randomized controlled trial, participants were assigned to the treatment condition (using an AI assistant, web search, and instructions) or the control condition (completing tasks with web search and instructions alone). The erosion of conceptual understanding, code reading, and debugging skills that we measured among participants using AI assistance suggests that workers acquiring new skills should be mindful of their reliance on AI during the learning process. Among participants who use AI, we find a stark divide in skill formation outcomes between high-scoring interaction patterns (65%-86% quiz

score) vs low-scoring interaction patterns (24%-39% quiz score). The high scorers only asked AI conceptual questions instead of code generation or asked for explanations to accompany generated code; these usage patterns demonstrate a high level of cognitive engagement.

Contrary to our initial hypothesis, we did not observe a significant performance boost in task completion in our main study. While using AI improved the average completion time of the task, the improvement in efficiency was not significant in our study, despite the AI Assistant being able to generate the complete code solution when prompted. Our qualitative analysis reveals that our finding is largely due to the heterogeneity in how participants decide to use AI during the task. There is a group of participants who relied on AI to generate all the code and never asked conceptual questions or for explanations. This group finished much faster than the control group (19.5 minutes vs 23 minutes), but this group only accounted for around 20% of the participants in the treatment group. Other participants in the AI group who asked a large number of queries (e.g., 15 queries), spent a long time composing queries (e.g., 10 minutes), or asked for follow-up explanations, raised the average task completion time. These contrasting patterns of AI usage suggest that accomplishing a task with new knowledge or skills does not necessarily lead to the same productive gains as tasks that require only existing knowledge.

Together, our results suggest that the aggressive incorporation of AI into the workplace can have negative impacts on the professional development workers if they do not remain cognitively engaged. Given time constraints and organizational pressures, junior developers or other professionals may rely on AI to complete tasks as fast as possible at the cost of real skill development. Furthermore, we found that the biggest difference in test scores is between the debugging questions. This suggests that as companies transition to more AI code writing with human supervision, humans may not possess the necessary skills to validate and debug AI-written code if their skill formation was inhibited by using AI in the first place.

## 7.1 Future Work

Our work is a first step to understanding the impact of AI assistance on humans in the human-AI collaboration process. We hope that this work will motivate future work that addresses the following limitations:

- **Task Selection:** This study focuses on a single task using a chat-based interface. This should be a lower bound for cognitive offloading since agentic AI coding tools would require even less human participation. In our work, users who relied on AI without thinking performed the worst on the evaluation; a completely agentic tool would create a similar effect. Future work should investigate the impacts of agentic coding tools on learning outcomes and skill development.
- **Task Length:** Ideally, skill formation takes place over months to years. We measured skill formation for a specific Python library over a one-hour period. Future work should study real-world skill development through longitudinal measurement of the impacts of AI adoption.
- **Participant Realism:** While participants in our study were professional or freelance programmers, there was not the same incentive to learn the library as if it were required for their actual job. Future studies should aim at studying the skill acquisition from novice workers within a real company.
- **Prompting Skills:** We collect self-reported familiarity with AI coding tools, but we do not actually measure differences in prompting techniques. An extension to our work would also involve testing the level of prompting fluency beyond self-report.
- **Evaluation Design:** Our study measures skill formation through a comprehensive quiz. Other studies could use the completion of another task or design coding as alternative evaluation strategies.
- **Human Assistance:** We do not include the counterfactual of how skill formation would be impacted by receiving assistance from humans. Since human assistance and feedback takes place in a diverse settings (e.g., classroom, pair programming, code review), future work can compare the effect of feedback from AI vs humans in all these settings on skill formation.

For novice workers in software engineering or any other industry, our study can be viewed as a small piece of evidence toward the value of intentional skill development despite the ubiquity of AI tools. Our study demonstrates the benefits of deploying cognitive effort when encountering a learning opportunity to master a

new tool even if barriers (e.g., errors) may be encountered in the process of mastery. Exerting cognitive effort can be assisted by AI; beyond the patterns we describe, major LLM services also provide learning modes (e.g., ChatGPT Study Mode, Claude Code Learning / Explanatory mode). Ultimately, to accommodate skill development in the presence of AI, there needs to be a more expansive view of the impacts of AI on workers. Participants in the new AI economy must care not only about productivity gains from AI but also the long-term sustainability of expertise development amid the proliferation of new AI tools.

